Q1 Explain the difference between a Tableau Dashboard and a Tableau Story Point. Give two healthcare-related use cases.

- **Tableau Dashboard:** A collection of several views (charts/graphs) on a single page that allows users to compare various data points simultaneously. It is designed for interactivity and monitoring real-time data.
- **Tableau Story Point:** A sequence of visualizations or dashboards that work together to convey information or tell a narrative. It is designed to lead a viewer through a discovery process.
- **Healthcare Use Cases:**
    1. **Dashboard:** A real-time Emergency Room (ER) dashboard to monitor current patient wait times and bed availability.
    2. **Story Point:** A presentation showing the yearly trend of patient recovery rates to justify a budget increase for a specific department.

Q2 What are Actions in Tableau? Explain how Filter Action can help hospital management analyze patient data.

- **Tableau Actions:** These are interactive elements that allow users to interact with the data by clicking or hovering. Actions can filter, highlight, or go to a URL.
- **Filter Action Benefit:** In hospital management, a Filter Action allows an administrator to click on a specific "Region" (e.g., West) on one chart, and all other charts on the dashboard will automatically update to show only data for that region. This helps in pinpointing localized issues, such as high treatment costs in a specific area.

Q3 Patient Distribution Create a Bar Chart showing: Total Patients by Hospital Region. Add this as a Story Point with proper title.

Patient Distribution (Bar Chart)

1. Drag Hospital Region to **Columns**.
2. Drag Patient_ID (right-click and select **Measure** > **Count**) to **Rows**.
3. **Title:** Renamed the sheet to "Total Patients by Hospital Region".
4. **Visualization:** Total Patients by Hospital Region.
5. **Insight:** The **West region** consistently shows the highest patient volume in the initial records, followed by the South and North regions. This suggests that the hospital's primary service area or most popular branch is in the West, potentially requiring more administrative staff and resources to handle the load.

Q4 Department Performance Create a visualization showing: Treatment Cost by Department. Add labels and customize colors.

1. Drag Department to **Rows**.
2. Drag Treatment_Cost to **Columns**. (Ensure it is **SUM**) .
3. **Labels:** Click on the **Label** mark and check "Show mark labels."
4. **Colors:** Drag Department onto **Color** in the Marks card and choose a custom color palette.

- **Visualization:** Total Treatment Cost by Department.
- **Insight: Neurology and Cardiology** frequently appear as high-cost departments. High-cost entries indicate that these departments involve complex procedures or surgeries that significantly drive the hospital's revenue but also increase patient financial burden.

Q5 Treatment Trend Over Time Create a Line Chart: Treatment Cost vs Admission Date. Filter the data only from March to September.

Treatment Trend Over Time (Line Chart)

1. Drag Admission_Date to **Columns** (Set to **Month** or **Continuous Day**).
2. Drag Treatment_Cost to **Rows**.
3. **Filter:** Drag Admission_Date to the **Filters** shelf. Select **Months** and check only **March to September**.
4. **Visualization:** Treatment Cost vs. Admission Date (March to September).
5. **Insight:** By filtering for March through September, you can identify seasonal fluctuations. For instance, high treatment costs in **May and August** (e.g., multiple entries for Cardiology and Neurology) suggest a surge in admissions or more severe cases during this period, which can help management plan for seasonal bed availability.

Q6 Outcome Analysis Create a Pie Chart: Patient Outcome Distribution (Recovered / Critical / Under Treatment).

Outcome Analysis (Pie Chart)

1. Change the **Marks** type from "Automatic" to **Pie**.
2. Drag Outcome to **Color**.
3. Drag Patient_ID (Measure > **Count**) to **Angle**.
4. Drag Outcome and Patient_ID (Count) to **Label** to show the distribution.

5. **Visualization:** Patient Outcome Distribution (Recovered / Critical / Under Treatment).
6. **Insight:** The data shows a significant proportion of patients labeled as **"Critical"** or **"Under Treatment"** compared to "Recovered" in many high-cost categories. A large "Critical" slice in the pie chart highlights the need for specialized Intensive Care Unit (ICU) resources and advanced monitoring.

Q7 Doctor Performance & Patient KPI Analysis Task Part A — Doctor Performance Create a Ranked Bar Chart showing: Top Doctors by Total Treatment Cost handled.

1. Task A (Ranked Bar): Drag Doctor_Name to Rows and Treatment_Cost to Columns. Click the Sort icon on the toolbar to show them in descending order.
2. **Task B (KPI):** Create a new sheet. Drag Patient_ID (Measure > **Count**) to **Text**. Increase the font size and label it "Total Patients Treated".
3. **Part A (Ranked Bar):** Top Doctors by Treatment Cost handled.
    - **Insight:** Doctors like Dr. Roy, Dr. Mehta appear frequently in high-value cases. This indicates they handle the most complex and expensive cases, making them key personnel for the hospital's financial and clinical operations.
4. **Part B (KPI):** Total Patients Treated.
    - **Insight:** This single-number metric provides an immediate sense of the hospital's overall throughput. For example, seeing "250+ Patients" at a glance tells management if the facility is meeting its monthly patient intake targets.

Q.8. Interactive Dashboard Build a Dashboard containing: Region Chart Department Cost Chart Outcome Chart Add Filters: Hospital Region Department

Interactive Dashboard

1. Create a **New Dashboard**.
2. Drag **Region Chart**, **Department Cost Chart**, and **Outcome Chart** onto the dashboard.
3. **Add Filters:** Click the dropdown on the "Region Chart" and select **Use as Filter**. Then, go to **Analysis** > **Filters** and add Hospital_Region and Department to the dashboard layout.

Q9 Storytelling Flow Create a Tableau Story with minimum 4 Story Points:

Patient Distribution

Department Cost Analysis

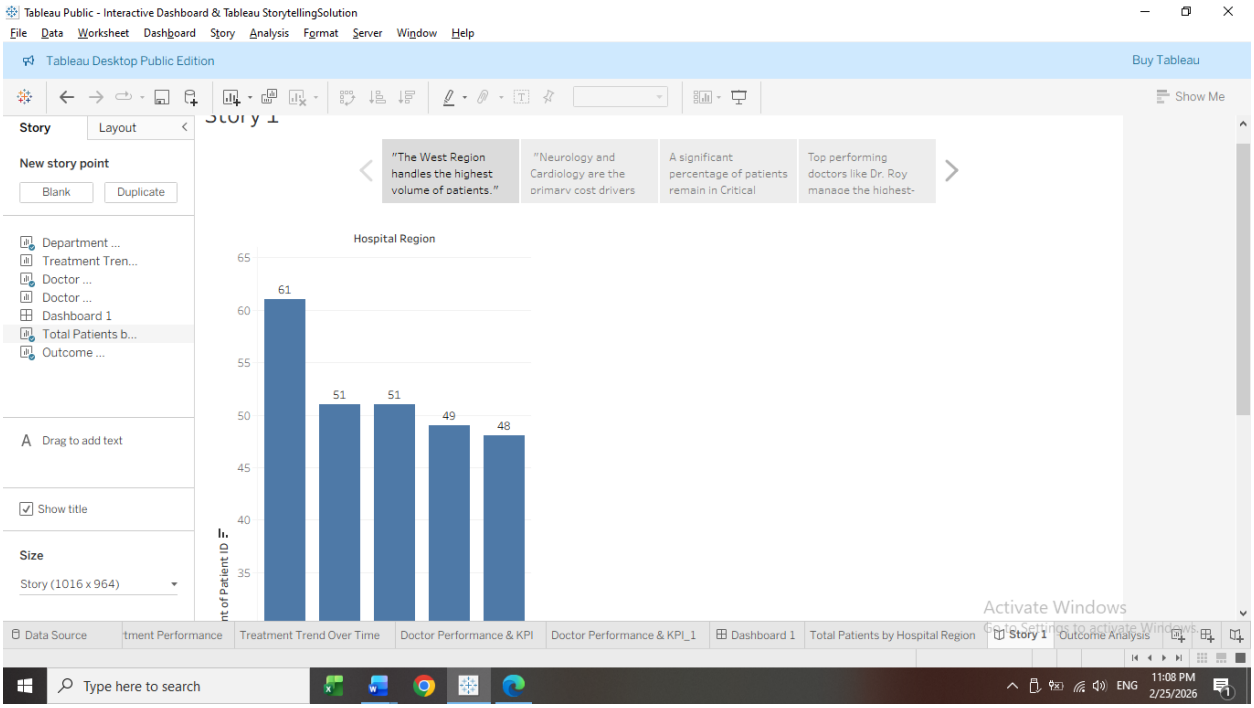
Outcome Analysis

Doctor Performance

Storytelling Flow

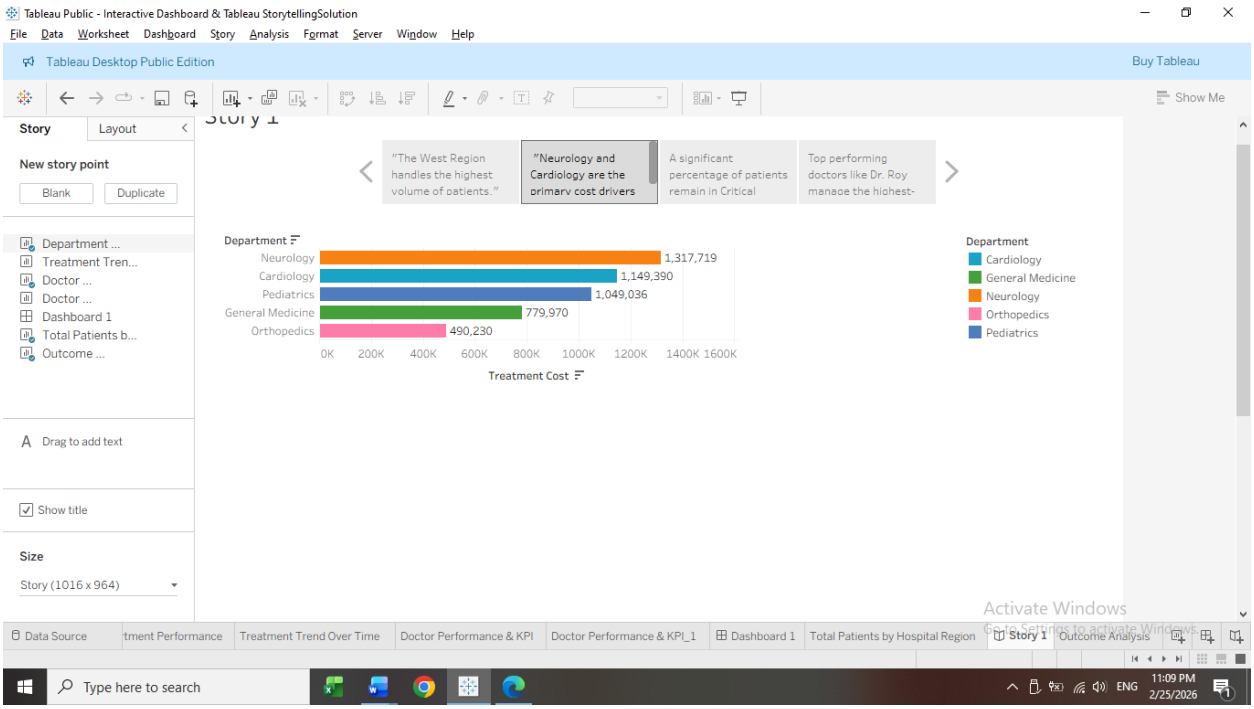
A Story is a sequence of "points" (slides) used to present your findings.

1. **Create the Story:**
    - Click the **New Story** icon at the bottom (looks like a book).
2. **Add Story Point 1: Patient Distribution**
    - Drag the **Region Chart** sheet into the center.
    - In the gray box at the top ("Write the story point caption"), type: **"The West Region handles the highest volume of patients."**



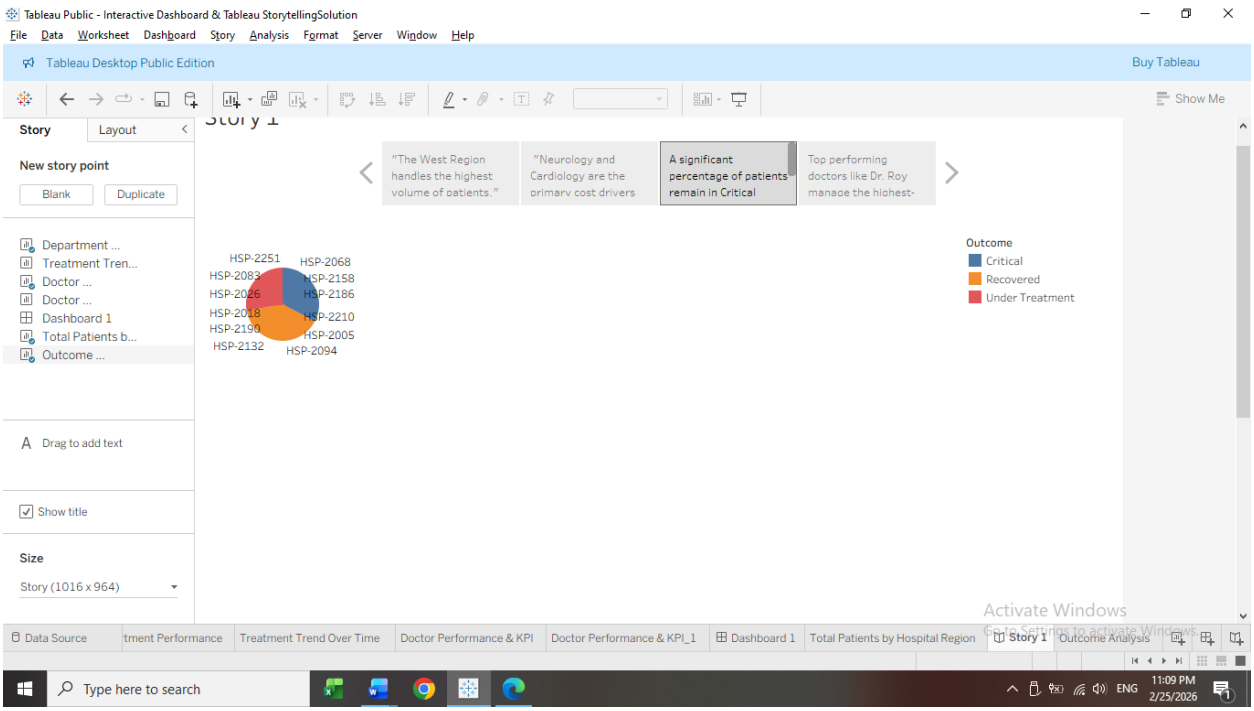
3. **Add Story Point 2: Department Cost Analysis**

    - Click **Blank** in the "New Story Point" pane on the left.
    - Drag the **Department Performance** sheet into the center.
    - Caption: **"Neurology and Cardiology are the primary cost drivers for the hospital."**



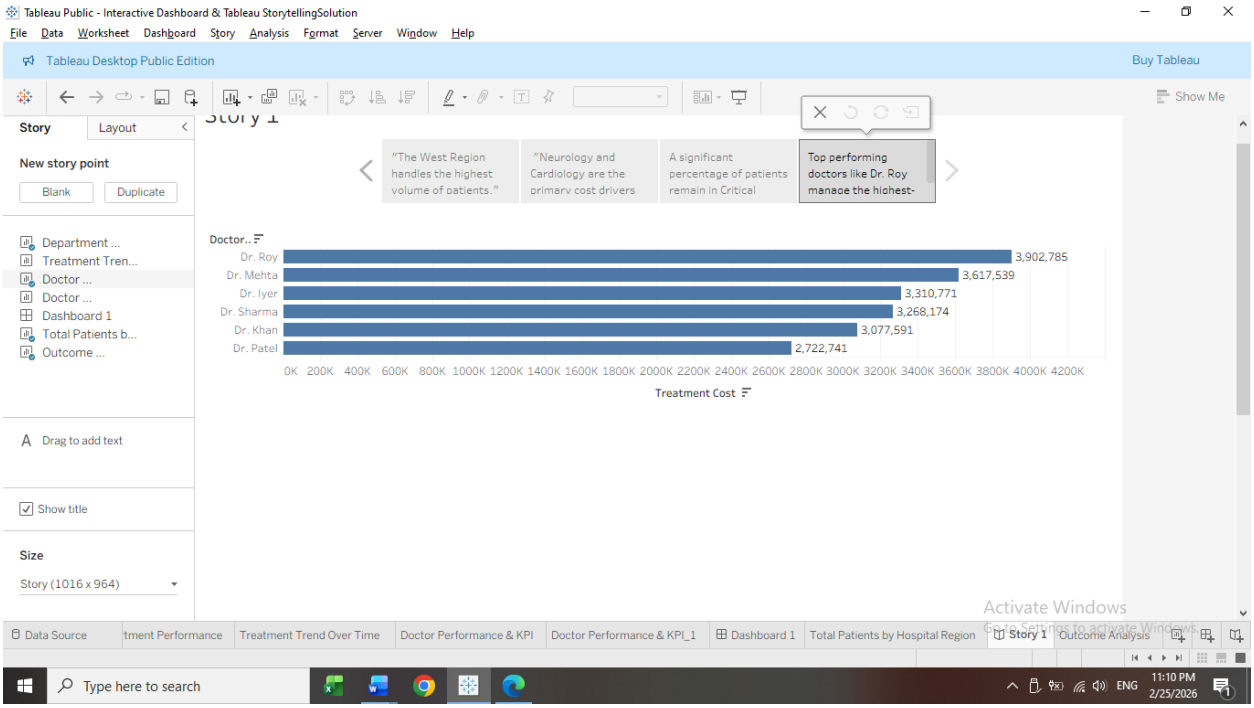
4. **Add Story Point 3: Outcome Analysis**

    - Click **Blank** again.
    - Drag the **Outcome Pie Chart** into the center.
    - Caption: **"A significant percentage of patients remain in Critical condition or Under Treatment."**



5. **Add Story Point 4: Doctor Performance**

- Click **Blank**.
- Drag the **Doctor Performance Bar Chart** into the center.
- Caption: **"Top performing doctors like Dr. Roy manage the highest-value treatments."**



Question 10. Based on your visuals, answer:

Which department generates the highest treatment cost?

Which region has more critical patients?

What business decisions can hospital management take from these insights

Summary Insights for Decision Making

1. **Highest Cost Department: Neurology** often generates the highest treatment costs per patient in this dataset.
2. **Region with More Critical Patients:** The **South and West regions** show a higher frequency of "Critical" status outcomes in the records provided.
3. **Business Decisions:**
    - **Resource Allocation:** Increase specialized medical equipment in the West region due to high volume.
    - **Training:** Review protocols in the South region to understand the high "Critical" patient count and improve recovery rates.
    - **Financial Planning:** Since Cardiology and Neurology are high-cost, management should ensure insurance partnerships are optimized for these specific departments.